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Start Hadoop :

- start-dfs.sh
- start-yarn.sh

Upload Dataset to HDFS :

```
hdfs dfs -mkdir /input
```

```
hdfs dfs -put forestfires.csv /input/
```

- check :

```
hdfs dfs -ls /input/
```

code :

- mapper.py

```
#!/usr/bin/env python3
```

```
import sys
```

```
for line in sys.stdin:
```

```
    line = line.strip()
```

```
# Skip header
if line.startswith("X"):
    continue

data = line.split(",")

month = data[2]
area = float(data[12])

print(f"{month}\\t{area}")
```

- reducer.py

```
#!/usr/bin/env python3
import sys

current_month = None
total_area = 0

for line in sys.stdin:
    line = line.strip()
    month, area = line.split("\t")
    area = float(area)
```

```
    if current_month == month:
        total_area += area
    else:
        if current_month:
            print(f"{current_month}\t\t{total_area}")
            current_month = month
            total_area = area
        if current_month:
            print(f"{current_month}\t\t{total_area}")
```

give permissions :

chmod +x mapper.py reducer.py

Open mapred-site.xml :

nano \$HADOOP_HOME/etc/hadoop/mapred-site.xml

Add this configuration

Paste inside `<configuration>` tag:

```
<property>
<name>mapreduce.framework.name</name>
<value>yarn</value>
</property>

<property>
<name>yarn.app.mapreduce.am.env</name>
```

```
<value>HADOOP_MAPRED_HOME=/root/hadoop</value>
</property>

<property>
<name>mapreduce.map.env</name>
<value>HADOOP_MAPRED_HOME=/root/hadoop</value>
</property>

<property>
<name>mapreduce.reduce.env</name>
<value>HADOOP_MAPRED_HOME=/root/hadoop</value>
</property>
```

Restart Hadoop :

[stop-dfs.sh](#)

[stop-yarn.sh](#)

[start-dfs.sh](#)

[start-yarn.sh](#)

Remove old output folder :

```
hdfs dfs -rm -r /output_fire
```

Run your job :

```
hadoop jar $HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming*.jar \
-input /input/forestfires.csv \
-output /output_fire \
-mapper mapper.py \
-reducer reducer.py \
-file mapper.py \
-file reducer.py
```

Check Output :

```
hdfs dfs -cat /output_fire/part-00000
```

